

# Why You Need Ethical Hacker Certification and 7 Options to Consider

## What Is Ethical Hacker Certification?

An Ethical Hacker Certification is a professional credential that verifies an individual's proficiency in ethical hacking, cybersecurity, and penetration testing. This certification showcases their ability to detect vulnerabilities in computer systems, networks, or applications and legally exploit them to improve an organization's security posture.

By obtaining this certification, individuals can demonstrate their expertise as ethical or "white-hat" hackers, who utilize the same tools and tactics as malicious hackers but with the intention of preventing cyberattacks rather than causing damage

## Advantages of Ethical Hacker Certification

Acquiring an ethical hacker certification can significantly influence your career as a security professional. This certification not only confirms your skills and knowledge in the field but also presents new job opportunities, boosts credibility, and raises earning potential.

## Expanded Job Opportunities

In the competitive job market of today, holding an ethical hacking certification distinguishes you from other candidates competing for similar positions. Many organizations prioritize hiring professionals with certifications to ensure they possess the required expertise to protect their systems from cyber threats. By obtaining a recognized ethical hacking credential, you demonstrate that you have both theoretical knowledge and practical experience in identifying vulnerabilities and securing networks.

## Enhanced Salary Prospects

Achieving a certified ethical hacking credential can result in higher pay, with [PayScale data](#) indicating that those who hold this qualification earn an average of \$91k per annum in the U.S., significantly more than non-certified individuals.

## Improved Security Knowledge

- **Core knowledge:** An ethical hacker certification offers comprehensive training on various cybersecurity aspects such as network security protocols, cryptography techniques, web application security testing methodologies, etc., which helps establish a strong foundation for further specialization within the field.
- **Penetration testing skills:** The certification process also includes learning how to conduct penetration testing, an essential skill for ethical hackers. Hands-on practice provides an opportunity to understand the thought process of malicious entities and identify any weak points in networks before they are exploited.
- **Staying up-to-date:** Cybersecurity threats are constantly evolving, and it's crucial for professionals to remain knowledgeable of the latest trends and techniques. Ethical hacker certifications often require recertification or continuing education credits, ensuring that certified individuals maintain their knowledge base and stay current on emerging security issues.

## Increased Credibility

Obtaining an ethical hacking certification demonstrates your dedication to the field of cybersecurity. It shows employers that you have invested time and effort in enhancing your skills. Additionally, having a recognized credential helps build trust among clients who may seek your expertise as a consultant or freelance professional.

## Essential Skills for Ethical Hacking

Aspiring ethical hackers must develop a robust foundation in various cybersecurity domains to excel in their careers. Here are a few competencies that are essential for ethical hackers to succeed, and are covered in most ethical hacker certifications.

### Network Security

Securing computer networks from unauthorized access, misuse, or damage is crucial for any organization's IT infrastructure, and an ethical hacker should be knowledgeable about networking protocols and devices such as routers and switches. Additionally, understanding firewalls and intrusion detection/prevention systems (IDS/IPS) is vital for securing an organization's network perimeter.

# Cryptography

Cryptography is essential in protecting sensitive information by encrypting it into unreadable formats that can only be deciphered with specific keys. A skilled ethical hacker should be proficient in cryptographic algorithms like AES (Advanced Encryption Standard), RSA (Rivest-Shamir-Adleman), and SHA (Secure Hash Algorithm). Understanding public key infrastructure (PKI) concepts, such as digital signatures and certificates, is also necessary for secure communication between parties over untrusted networks.

## Penetration Testing

Penetration testing, also known as pen testing or ethical hacking, involves simulating real-world cyberattacks on an organization's IT assets to identify vulnerabilities before malicious hackers exploit them. An effective pen tester needs expertise in:

- **Reconnaissance techniques** for gathering information about the target.
- **Vulnerability scanning tools** to detect potential weaknesses.
- **Exploitation frameworks** such as Metasploit to launch simulated attacks on discovered vulnerabilities.
- **Post-exploitation activities**, including privilege escalation, lateral movement, and maintaining access within compromised systems.

## Web Application Security

Web application security is a critical aspect of ethical hacking since web applications are often targeted by cybercriminals. Familiarity with common web vulnerabilities like SQL injection, cross-site scripting (XSS), and broken authentication is essential. Additionally, knowledge of secure coding practices in languages like PHP, Java, or Python can help you better understand how these vulnerabilities arise and recommend appropriate remediation measures.

## Social Engineering Techniques

Social engineering involves manipulating individuals into revealing sensitive information or performing actions that compromise their security. Ethical hackers must be aware of social engineering tactics such as phishing emails, pretexting calls, baiting USB drives, etc., to educate employees about potential threats and design effective countermeasures against them.

## Top Ethical Hacking Certifications

Here is a list of some of the most important ethical hacking certifications.

## 1. CEH (Certified Ethical Hacker)

The [Certified Ethical Hacker](#) certification, provided by the EC-Council, is one of the most well-known and globally recognized ethical hacking certifications. CEH covers various topics such as reconnaissance, network scanning, vulnerability analysis, system hacking techniques, social engineering attacks, and web application vulnerability assessments, among others.

## 2. S-EHP (SECO Ethical Hacking Practitioner)

The [SECO Ethical Hacking Practitioner](#) certification offers an in-depth understanding of penetration testing methods, tools, and techniques employed by ethical hackers to identify vulnerabilities in systems and networks. This certification is ideal for security professionals aiming to enhance their offensive security skills.

## 3. GPEN (GIAC Penetration Tester)

The [GIAC Penetration Tester](#) certification concentrates on evaluating network security using manual and automated methods while adhering to industry best practices, such as the OWASP Top Ten Project guidelines during assessments.

## 4. CHFI (Computer Hacking Forensic Investigator)

The [Computer Hacking Forensic Investigator](#) certification, also provided by the EC-Council, is tailored for professionals who want to specialize in digital forensics and incident response. This certification covers various topics such as data acquisition, evidence preservation, and analysis of volatile and non-volatile data, among others.

## 5. OSCP (Offensive Security Certified Professional)

The [Offensive Security Certified Professional](#) is a highly esteemed ethical hacking certification that emphasizes hands-on penetration testing skills. OSCP candidates must successfully complete a 24-hour practical exam in which they must compromise multiple systems within the allotted time.

## 6. CSTA (Certified Security Testing Associate)

The [Certified Security Testing Associate](#) certification aims to provide a fundamental understanding of security testing methodologies and tools employed by ethical hackers during vulnerability assessments or penetration tests of web applications and networks.

## 7. CompTIA PenTest+

The [CompTIA PenTest+](#) certification, another popular choice among cybersecurity professionals, covers both manual and automated techniques for assessing network vulnerabilities. It also focuses on management aspects such as project scop

## Ethical Hacker Certification with HackerOne

HackerOne has been the leader in providing access to the largest, most diverse community of ethical hackers in the world. The most demanding global brands and government agencies rely on HackerOne to maximize their relationships with the hacking community to achieve attack resistance and continuously protect their attack surface.

With a portfolio that spans Attack Surface Management, Pentest as a Service, Bug Bounty, and Vulnerability Disclosure Programs, HackerOne helps customers outsmart cybercriminals by tapping into the security expertise of the ethical hacker community.

[Learn more about HackerOne](#)

Company

Leadership

Careers

Partners

Newsroom

Contact Us

Knowledge

Center

Application Security

Penetration Testing

Cloud Security

Hacking

Cybersecurity

Attacks

DevSecOps

Resources

Blog

Documentation

Leaderboard

Partner Portal

Resources

Contacted  
by a  
hacker?

